

We therefore only need to show that σ intertwines the differential, or equivalently,

$$\sigma(\mathbf{r}(t_1 + z - t_2)) = \mathbf{r}(t_1 - z + t_2). \quad (4.46)$$

This follows from applying τ_z to equation (4.20), as well as the fact that $\mathbf{r}$ is difference-dependent. $\square$

Remark 4.6. The commutativity condition of Proposition 4.5 will be called weak commutativity.

From Proposition 4.5, we can define a “meromorphic” tensor product $M_z \otimes N_0$ for a pair of FSF modules of $U_h(\mathfrak{d}(\mathcal{O}))$. However, the resulting module is not FSF anymore. Nevertheless, there is still a natural generalization of associativity condition as follows. Consider now a triple of FSF modules $M = U[[\hbar]]$, $N = V[[\hbar]]$ and $P = W[[\hbar]]$, the tensor product $M_{z+w} \otimes (N_w \otimes P_0)$ is naturally identified with the vector space

$$U \otimes V \otimes W[[\hbar]]((w))((z+w)) \quad (4.47)$$

where as $(M_z \otimes N)_w \otimes P_0$ is identified with

$$U \otimes V \otimes W[[\hbar]]((z))((w)). \quad (4.48)$$

These two are not immediately comparable, but they both receive a map from the vector space

$$U \otimes V \otimes W[[\hbar]][[z, w]][z^{-1}, w^{-1}, (z+w)^{-1}] \quad (4.49)$$

via formal expansion, just as in [FBZ04, Chapter 3]. We claim the following result, which is mostly clear from the definition.

Proposition 4.7. *For a triple of FSF modules $M = U[[\hbar]]$, $N = V[[\hbar]]$ and $P = W[[\hbar]]$, the following statements are true.*

- *The vector space*

$$U \otimes V \otimes W[[\hbar]][[z, w]][z^{-1}, w^{-1}, (z+w)^{-1}] \quad (4.50)$$

has a well-defined DG module structure of $U_h(\mathfrak{d}(\mathcal{O}))$, whose action is induced by $(1 \otimes \Delta_w)\Delta_{z+w} = (\Delta_z \otimes 1)\Delta_w$, and whose differential is given by

$$d_{\mathbf{r}}(z, w) := d_M + d_N + d_P - 2\hbar(\mathbf{r}^{12}(t_1 + z - t_2) + \mathbf{r}^{13}(t_1 + z + w - t_2) + \mathbf{r}^{23}(t_1 + w - t_2)). \quad (4.51)$$

We denote this module by $M_{z+w} \otimes N_w \otimes P_0$, signifying its independence of order.

- *Formal expansion as discussed above gives the following maps of DG modules of $U_h(\mathfrak{d}(\mathcal{O}))$, both of which are equivalences after appropriate ring extension:*

$$\begin{array}{ccc} & M_{z+w} \otimes N_w \otimes P_0 & \\ \swarrow & & \searrow \\ (M_z \otimes N)_w \otimes P_0 & & M_{z+w} \otimes (N_w \otimes P_0) \end{array} \quad (4.52)$$